



Peak Achievers Online Academy

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MATHEMATICS

GRADE 7

BOOK 1

Comprehensive Revision Notes, Worked Examples,
Topical Questions, Mixed Exercises and Final Tests



HINT

REMEMBER

THINK BIG

SPOTLIGHT

Mathematics for a Brighter Future

Editorial Audit Note

This edition has been strengthened for Peak Achievers use with HINT, REMEMBER, SPOTLIGHT, NB and THINK BIG callouts, application challenges, corrected curriculum-scope gaps, non-repetitive mixed exercises and clearer final-test questions. The strand and sub-strand map was checked against the KICD Grade 7 Mathematics curriculum design.

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5.1 Data Handling

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APPLICATION QUESTION

APPLICATION CHALLENGE: Collect the number of minutes five learners take to travel to school. Make a frequency table, choose a suitable representation, and explain one conclusion from the data.

STRAND 1 - NUMBERS**1.1 Whole Numbers**

Whole numbers are 0, 1, 2, 3, Their value depends on the position of each digit.

Place value tells the value of a digit according to its position; face value is the digit itself.

Expanded form separates a number into place values. Example: $48,756 = 40,000 + 8,000 + 700 + 50 + 6$.

Compare whole numbers using $>$, $<$ and $=$. Arrange numbers by comparing digits from the highest place value.

Rounding gives a convenient approximate value. Check the digit immediately to the right of the required place.

The four basic operations are addition, subtraction, multiplication and division.

Use BODMAS when an expression contains several operations.

Estimation is useful for checking whether an answer is reasonable. Natural numbers can also be classified as even, odd or prime, and number sequences can be identified and extended using a rule.

REMEMBER: Show working clearly and check whether your answer is reasonable.

REMEMBER: Large numbers are easiest to read by separating them into groups of three digits from the right.

HINT: When rounding, look only at the digit immediately to the right of the required place.

SPOTLIGHT: Whole numbers appear in population figures, budgets, enrolment records, stock counts and distances.

NB: Do not just calculate - explain why your method works and whether the answer is sensible.

THINK BIG: A school receives 248,750 exercise books and distributes 196,485. Estimate first, then calculate the exact number remaining and explain whether your answer is reasonable.

Worked examples

$$48,756 + 23,489 = 72,245$$

$$75,000 - 28,475 = 46,525$$

$$48 \times 25 = 1,200$$

$$2,400 \div 30 = 80$$

$$18 + 6 \times 4 = 42$$

$$35 \times 48 = 1,680; \text{ if } 1,275 \text{ items are used, } 405 \text{ remain.}$$

Revision Quiz

1. Write 506,072 in words.
2. State the place value of 8 in 483,615.
3. Write 704,305 in expanded form.
4. Round 86,749 to the nearest thousand.
5. Evaluate $36 + 8 \times 4$.
6. Evaluate $96 - 48 \div 8 \times 5 + 14$.
7. A school has 1,275 boys and 1,485 girls. Find the total population.
8. Estimate $4,982 + 3,071$ to the nearest thousand.

APPLICATION QUESTION

APPLICATION CHALLENGE: A school receives 248,750 exercise books and distributes 196,485. Estimate first, then calculate the exact number remaining and explain whether your answer is reasonable.

1.2 Factors

A factor divides a number exactly. Factor pairs can be used to list all factors.

A prime number has exactly two factors: 1 and itself. A composite number has more than two factors. The number 1 is neither prime nor composite.

Divisibility rules help test numbers quickly: 2 (even), 3 (digit sum divisible by 3), 5 (ends 0 or 5), 9 (digit sum divisible by 9), 10 (ends 0).

The HCF is the greatest factor common to two or more numbers.

Prime factorisation expresses a number as a product of prime factors.

REMEMBER: Show working clearly and check whether your answer is reasonable.

REMEMBER: A factor divides exactly with no remainder.

HINT: Use divisibility tests before attempting full division.

SPOTLIGHT: HCF is useful when making the greatest possible number of identical groups.

NB: Do not just calculate - explain why your method works and whether the answer is sensible.

THINK BIG: A farmer has 84 avocado seedlings and 126 mango seedlings. Find the greatest number of identical groups that can be made and state the composition of each group.

Worked examples

Factors of 24: 1, 2, 3, 4, 6, 8, 12, 24.

HCF of 18 and 24 = 6.

$$60 = 2^2 \times 3 \times 5.$$

$$72 = 2^3 \times 3^2.$$

HCF of 48 and 72 = 24.

Revision Quiz

1. List all factors of 36.
2. State whether 29 is prime or composite.
3. Find the HCF of 20 and 30.
4. Express 120 as a product of prime factors.
5. Find the HCF of 60 and 84.
6. 36 orange and 48 mango seedlings are put into identical groups. Find the greatest possible number of groups.

APPLICATION QUESTION

APPLICATION CHALLENGE: A farmer has 84 avocado seedlings and 126 mango seedlings. Find the greatest number of identical groups that can be made and state the composition of each group.

1.3 Fractions

A fraction represents part of a whole and has a numerator and denominator.

Fractions may be proper, improper or mixed numbers. Convert between improper fractions and mixed numbers when useful.

Equivalent fractions have the same value. Simplify by dividing numerator and denominator by a common factor.

For addition and subtraction, use a common denominator. For multiplication, multiply numerators and denominators then simplify. For division, multiply by the reciprocal of the divisor.

Fractions are widely used in sharing, measurements, recipes, money and quantities.

REMEMBER: Show working clearly and check whether your answer is reasonable.

REMEMBER: Always simplify a final fraction where possible.

HINT: For addition or subtraction, first obtain a common denominator.

SPOTLIGHT: Fractions model sharing, recipes, land portions, water use and money.

NB: Do not just calculate - explain why your method works and whether the answer is sensible.

THINK BIG: A 1,800 L tank is $\frac{5}{6}$ full. A school uses $\frac{2}{5}$ of the water present. How many litres remain?

Worked examples

$$2\frac{3}{4} = 11/4.$$

$$17/5 = 3\frac{2}{5}.$$

$$18/24 = 3/4.$$

$$1/2 + 1/3 = 5/6.$$

$$5/6 - 1/4 = 7/12.$$

$$2\frac{1}{4} + 1\frac{1}{2} = 3\frac{3}{4}.$$

$$5\frac{1}{4} - 2\frac{2}{3} = 2\frac{7}{12}.$$

$$3/4 \times 2/5 = 6/20 = 3/10.$$

$$3/4 \div 2/5 = 15/8 = 1\frac{7}{8}.$$

$$3/5 \text{ of } 200 = 120.$$

Revision Quiz

1. Simplify $30/45$.
2. Convert 3% to an improper fraction.
3. Arrange $1/2$, $3/4$ and $2/3$ in ascending order.
4. Evaluate $5/6 - 1/4$.
5. Evaluate $7/8 \div 7/12$.
6. Find $3/8$ of 2,400.
7. A tank has 1,200 L. One quarter is used in the morning, then one third of the remainder is used in the afternoon. Find the amount left.

APPLICATION QUESTION

APPLICATION CHALLENGE: A 1,800 L tank is $5/6$ full. A school uses $2/5$ of the water present. How many litres remain?

1.4 Decimals

A decimal represents parts of a whole using place values to the right of the decimal point.

Compare decimals by aligning decimal points and comparing corresponding place values.

For addition and subtraction, align decimal points. For multiplication and division, use place value carefully.

Multiplying or dividing by 10, 100 or 1,000 shifts place values.

Decimals can be converted to fractions and rounded to required decimal places.

REMEMBER: Show working clearly and check whether your answer is reasonable.

REMEMBER: Align decimal points when adding or subtracting.

HINT: Estimate the answer before calculating to catch misplaced decimal points.

SPOTLIGHT: Decimals are used in money, mass, length and capacity.

NB: Do not just calculate - explain why your method works and whether the answer is sensible.

THINK BIG: A shopkeeper sells 2.75 kg of rice at KSh 168.50 per kilogram. Estimate the cost, then calculate the exact amount.

Worked examples

$$47.386 = 40 + 7 + 0.3 + 0.08 + 0.006.$$

$$3.45 < 3.50.$$

$$2.05 < 2.5 < 2.505 < 2.55.$$

$$24.75 + 8.60 = 33.35.$$

$$18.50 - 7.86 = 10.64.$$

$$3.6 \times 4 = 14.4.$$

$$2.5 \times 1.2 = 3.$$

$$8.4 \div 4 = 2.1.$$

Revision Quiz

1. Write 6.407 in expanded form.
2. Arrange 3.08, 3.8, 3.18 and 3.018 in ascending order.
3. Calculate $12.75 + 8.6 - 5.85$.
4. Convert 4.25 km to metres.
5. A book costs KSh 175.50, a pen KSh 32.75 and a ruler KSh 24.25. Find the change from KSh 300.

APPLICATION QUESTION

APPLICATION CHALLENGE: A shopkeeper sells 2.75 kg of rice at KSh 168.50 per kilogram. Estimate the cost, then calculate the exact amount.

1.5 Squares and Square Roots

The square of a number is obtained by multiplying the number by itself. Example: $8^2 = 64$.

A square root reverses squaring. Example: $\sqrt{144} = 12$.

Perfect squares include 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121 and 144.

Prime-factor pairs can be used to find square roots of perfect squares.

Squares and square roots are useful in area and in the Pythagorean relationship. Grade 7 work also includes squares and square roots of perfect-square fractions and decimals, for example $(0.4)^2 = 0.16$ and $\sqrt{0.81} = 0.9$.

REMEMBER: Show working clearly and check whether your answer is reasonable.

REMEMBER: Squaring and finding a square root are inverse operations.

HINT: For perfect-square fractions, square-root the numerator and denominator separately after simplifying.

SPOTLIGHT: Squares and square roots help determine dimensions of square spaces and support Pythagoras.

NB: Do not just calculate - explain why your method works and whether the answer is sensible.

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